# The Euclidean algorithm is not optimal among uniform processes

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**Abstract**

Reusing a stored Euclidean prefix, we derive $\gcd(a, b)$ from $a, b$ with $O(\log a / \log\log a)$ applications of the remainder function alone, uniformly for $a > b \geq 1$; Euclid's algorithm needs order $\log a$ in the worst case. These derivations bound the intrinsic complexity of gcd and coprimeness by the same order, refuting Moschovakis's conjectured logarithmic lower bounds for uniform processes; by his best-uniform-process theorem, a single uniform process attains this bound. The Main Conjecture for algorithms remains open. We also prove an $\Omega(\log\log a)$ lower bound for almost all coprime pairs and give, for each $\varepsilon > 0$, a nondeterministic gcd program with a fixed number of registers and no recursion that makes at most $\varepsilon \log a + O(1)$ remainder calls. The main bounds and the refutations are formalised in Lean 4; the best-uniform-process theorem is cited.

**AI transparency statement.** This paper was produced entirely by artificial intelligence, including the identification of the main result, its formalisation in Lean, and the writing of the manuscript. The human author initiated the broader research and directed the AI systems, but did not write the text. Details are given at the end of the paper.

## 1 Introduction

Van den Dries and Moschovakis asked whether the Euclidean algorithm is optimal among its peers, the algorithms that compute the gcd from the remainder function [9, p. 390]. Moschovakis's Main Conjecture asserts that for every algorithm $\alpha$ from rem and the zero test $\mathrm{eq}_0$ computing $\gcd(x, y)$ there is an $r > 0$ such that $\alpha$ makes more than $r \log x$ remainder calls on infinitely many pairs $x > y \geq 1$ [6, p. 2]. We refute its formulations for uniform processes, the book's more general model (Section 5); for algorithms it remains open. Throughout, "the book" means Moschovakis's monograph [6], and logarithms are to base 2.

**Derivations.** A *derivation* from $a, b$ is a finite list that starts with $a, b$ and in which each further entry is $\mathrm{rem}(x, y)$ for earlier entries $x, y$ with $y > 0$. Each further entry is one *instruction*, that is, one remainder call. Entries already in the list are *available* and may be reused; no constants and no other operations are supplied. Let $G(a, b)$ be the least number of instructions in a derivation containing $\gcd(a, b)$, and let $D(a, b) = G(a, b)$ for coprime inputs. Thus $G(a, 0) = 0$ and $D(a, 1) = 0$. Arithmetic used in the proofs, such as subtraction, sorting or comparison, only selects instructions; it is never itself an instruction. Thus $G$ measures the length of a derivation, not the work of finding one.

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**Theorem 1.** *There is a constant $C$ such that for all integers $a \geq 4$ and $0 \leq b \leq a$,*

$$G(a,b) \leq C \frac{\log a}{\log \log a}.$$

*In particular this bound holds for $D(a,b)$ when $\gcd(a,b) = 1$.*

**Idea of the proof.** For sufficiently large $n$, let $a > b$ be coprime with $a \leq N = 2^n$. Store a Euclidean prefix $r_0 = a, r_1 = b, \ldots, r_h$, where $h = \lceil \sqrt{n} \rceil$. If the prefix reaches a positive remainder at most $N^{3/4}$, take one further remainder and recurse on the resulting smaller pair. Otherwise its consecutive pairs stay distinct modulo every $m > N^{1/3}$, so the prefix occupies at least $\sqrt{h}$ residue classes modulo $m$ (Lemma 3). A close pair of residues lets three remainder calls produce a positive value below $m/(\sqrt{h} - 1)$ (Lemma 2). Reusing the same prefix, we repeatedly shrink the modulus from $r_h$ to at most $N^{1/3}$ in $O(n/\log n)$ calls. Two recursive gcd derivations on inputs at most $N^{1/3}$ finish. If $T(n)$ is the worst derivation cost for inputs at most $2^n$, the resulting recurrence (5) gives $T(n) = O(n/\log n)$. Tuning the prefix length sharpens the upper coefficient to $6 + o(1)$ (Corollary 4).

**Consequences and further results.** Uniform processes are the book's abstract model of computation from given primitives; every deterministic or nondeterministic recursive program induces one [6, §4A, Proposition 4A.1]. Section 5 converts the derivations into small certificates. This bounds the intrinsic complexity of gcd and coprimeness, refutes Conjecture 3.4 of [5] and version (4F-5) of the Main Conjecture, and limits what the homomorphism method can prove. The book's best-uniform-process theorem then gives a single process with the same bound on every input. The upper bound also settles an open problem of van den Dries and Moschovakis on the closure depth of the gcd (Corollary 5).

We also prove that $D(a,b) \geq (1 - o(1)) \log \log N$ for all but $o(N^2)$ coprime pairs $2 \leq b < a \leq N$ (Proposition 11). Finally, fixing the prefix length gives, for each $\varepsilon > 0$, a nondeterministic register program, with no recursion, whose cost is at most $\varepsilon \log a + B$ on all inputs (Theorem 12). The number of registers grows as $\varepsilon$ decreases.

**Related work.** The known intrinsic lower bounds give $\Omega(\log \log a)$ on special families of coprime pairs [9, 10], [6, Theorem 6C.5], also for richer primitives such as addition and integer quotient, and for depth. Pratt's unpublished nondeterministic algorithm [6, Theorem 2E.2 and Problems x2E.17–x2E.19] matches this order on successive Fibonacci numbers [6, Corollary 6C.7]. For remainder-only derivations, our almost-all bound gives the same order on a set of full density among coprime pairs.

In a different model, Bshouty obtained the optimal bound $\Theta(\log N/\log \log N)$ for gcd on inputs in $[0, N]$ using a random access machine with linear operations, shifts, comparisons and indirect addressing [1, Theorems i and ii, p. 3]. His construction tabulates blocks of the binary gcd algorithm [8], [3, §4.5.2], computing table entries on demand and counting their construction [1, §4, Algorithm D and Lemma 5, pp. 12–14]. Here the saving comes from reusing a Euclidean prefix with remainder as the only instruction.

## 2 Normalisation

Write $g = \gcd(a,b) > 0$. For positive $y$,

$$\operatorname{rem}(gx, gy) = g \operatorname{rem}(x, y).$$

Every generated value is divisible by $g$. Consequently scaling a derivation, or dividing all its values by $g$, preserves its length and the validity of every instruction. Hence

$$G(a,b) = D(a/g, b/g). \tag{1}$$

Define

$$T(n) = \max_{0\le b\le a\le 2^n,\ a>0} G(a,b).$$

Euclid's remainders form a derivation containing $\gcd(a,b)$, so $G$ and $T$ are finite. By (1) it suffices to bound $D$; the cases $a=b$, $b=0$ and $b=1$ need no instructions.

## 3 Three-call contractions

**Lemma 2.** *If $m>0$ is available and a finite collection $S$ of available values occupies $t\ge 2$ residue classes modulo $m$, at most three instructions generate $w$ with*

$$0 < w < \frac{m}{t-1}.$$

*Proof.* Sort the distinct residues. Some adjacent pair $u<v$ has gap $\delta = v-u < m/(t-1)$, because the sum of the $t-1$ adjacent gaps is less than $m$. Each selected residue is $\mathrm{rem}(x,m)$ for some $x\in S$, so it costs one instruction. If $u=0$, use $v$. If $0<u\le\delta$, use $u$. Otherwise $0<\delta<u$ and $v=u+\delta$, so $\mathrm{rem}(v,u)=\delta$. This uses at most three instructions. □

**Lemma 3** (Euclidean-prefix lemma)**.** *Let $r_0=a>r_1=b>\cdots>r_h>0$ be a Euclidean prefix of a coprime pair, with $r_{i-1}=q_ir_i+r_{i+1}$ and $q_i\ge 1$ for $1\le i<h$. For every integer $m>2a/r_h$, the $h$ pairs*

$$(r_i \bmod m, r_{i+1} \bmod m), \qquad 0\le i<h,$$

*are distinct. The prefix therefore occupies $t$ residue classes with $t^2\ge h$.*

*Proof.* Suppose the pairs at $i<j$ coincide. Set

$$M = \prod_{\ell=i+1}^{j} \begin{pmatrix} q_\ell & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} A & B \\ C & E \end{pmatrix}, \qquad \begin{pmatrix} r_i \\ r_{i+1} \end{pmatrix} = M \begin{pmatrix} r_j \\ r_{j+1} \end{pmatrix}.$$

Right multiplication by $\left(\begin{smallmatrix} q & 1 \\ 1 & 0 \end{smallmatrix}\right)$ sends $(A,B,C,E)$ to $(qA+B,\ A,\ qC+E,\ C)$, starting from $(q,1,1,0)$ for a single factor. By induction, all entries are nonnegative, $A\ge B\ge E$, $A\ge C\ge 1$, $B\ge 1$, and $\det M = (-1)^{j-i}$. Also $Ar_j\le r_i\le a$ and $r_j>r_h$ (as $j<h$), whence $A<a/r_h$. Write

$$\Delta = \det(M-I) = 1+\det M - A - E.$$

For an odd number of factors, $\Delta = -A-E<0$. By the same recursion, a product of at least two factors has $A\ge 2$ and $E\ge 1$; thus for an even positive number of factors, $\Delta = 2-A-E<0$. In either case,

$$0<|\Delta|\le A+E\le 2A\le 2a/r_h<m.$$

The presumed congruence gives $(M-I)(r_j,r_{j+1})^{\mathsf{T}}\equiv 0 \pmod m$. Multiplication by the adjugate shows that $m$ divides both $\Delta r_j$ and $\Delta r_{j+1}$. Consecutive remainders are coprime, so Bézout's identity implies $m\mid\Delta$, a contradiction. Finally, $t$ individual residue classes give at most $t^2$ ordered pairs. □

Since Lemma 3 holds for every $m > 2a/r_h$, one stored prefix serves every later modulus, although these moduli are not known in advance. When $\sqrt{h} > 1$, Lemma 2 applied to the prefix gives three instructions that generate

$$0 < w < \frac{m}{\sqrt{h} - 1}. \tag{2}$$

We call these three instructions a *contraction* of $m$, and $w$ the new modulus.

For example, from $44, 17$ one instruction stores the prefix $44, 17, 10$. Modulo $m = 10 > 2 \cdot 44/10$ its residues are $4, 7, 0$, and the close pair $4 < 7$ gives the contraction

$$\mathrm{rem}(44, 10) = 4, \qquad \mathrm{rem}(17, 10) = 7, \qquad \mathrm{rem}(7, 4) = 3,$$

with new modulus $3 < 10/(3 - 1)$, as in Lemma 2. One more instruction, $\mathrm{rem}(10, 3) = 1$, finishes. This derivation takes five instructions; Euclid reaches 1 in four. The example illustrates contraction without a saving on this pair.

## 4 Construction and recurrence

Fix an integer $n > 16$ and put $N = 2^n$ and $h = \lceil \sqrt{n} \rceil > 4$. After normalisation, assume $a > b > 1$ are coprime and $a \le N$. We use $N^{3/4}$ as the *early threshold* and $N^{1/3}$ as the *target modulus*. Any thresholds $N^\alpha, N^\beta$ with $\alpha < 1$, $\alpha + \beta > 1$ and $2\beta < 1$ would do: these make the early recursive problem smaller, let the prefix lemma apply above the target, and make the two final problems together smaller. We take $\alpha = 3/4$ and $\beta = 1/3$.

**Phase (i): the prefix.** Starting from $r_1 = b$, generate the Euclidean remainders $r_2, r_3, \ldots$, one instruction each, until the most recent value is at most $N^{3/4}$ or $r_h$ is available. This costs at most $h$ instructions. Since the inputs are coprime, Euclid's algorithm reaches 1 before 0, so the stopping value is positive.

If the stopping value $c$ is at most $N^{3/4}$, generate the next remainder $s < c$ with one more instruction. The pair $(c, s)$ is coprime, with both entries at most $2^{\lceil 3n/4 \rceil}$. This gives the bound

$$h + 1 + T(\lceil 3n/4 \rceil). \tag{3}$$

**Phase (ii): contractions.** Otherwise $r_h > N^{3/4}$, so $2a/r_h < 2N^{1/4} < N^{1/3}$ (as $n > 12$), and Lemma 3 applies to every available $m > N^{1/3}$. Starting at $m = r_h$, repeatedly replace $m$ by the value $w$ of (2) until $0 < m \le N^{1/3}$. Set

$$\kappa = \sqrt{h} - 1 > 1, \qquad J = \left\lceil \frac{2n/3}{\log \kappa} \right\rceil.$$

There are at most $J$ contractions: if $J$ have occurred then $m < N/\kappa^J \le N^{1/3}$. Since $\log \kappa = \frac{1}{4} \log n + O(1)$, their cost is $3J = O(n/\log n)$. This is the source of the saving: the prefix costs only $O(\sqrt{n})$ instructions, and each contraction reduces $\log m$ by at least $\log \kappa = \frac{1}{4} \log n + O(1)$.

**Phase (iii): two recursive derivations.** The small $m$ need not be coprime to either input. Generate $u = \mathrm{rem}(a, m)$ and $v = \mathrm{rem}(b, m)$, then append a derivation of $d = \gcd(m, u)$ from $m, u$ and one of $e = \gcd(d, v)$ from $d, v$. Since $m \le 2^{\lceil n/3 \rceil}$ and $G$ is symmetric, each costs at most $T(\lceil n/3 \rceil)$. Finally

$$e = \gcd(\gcd(m, \mathrm{rem}(a, m)), \mathrm{rem}(b, m)) = \gcd(m, \gcd(a, b)) = 1.$$

The cost in this case is at most

$$h + 3J + 2 + 2T(\lceil n/3 \rceil). \tag{4}$$

**The recurrence.** The two case estimates (3) and (4) and normalisation imply

$$T(n) \le \max\{T(\lceil 3n/4 \rceil) + O(\sqrt{n}),\ 2T(\lceil n/3 \rceil) + O(n/\log n)\}. \tag{5}$$

All implied constants are independent of the inputs.

*Proof of Theorem 1.* For $n \ge 2$ let $f(n) = n/\log n$. The ratios

$$\frac{f(\lceil 3n/4 \rceil)}{f(n)} \longrightarrow \frac{3}{4}, \qquad \frac{2f(\lceil n/3 \rceil)}{f(n)} \longrightarrow \frac{2}{3}$$

show that, beyond a fixed threshold, $f(\lceil 3n/4 \rceil) \le \frac{7}{8} f(n)$ and $2f(\lceil n/3 \rceil) \le \frac{5}{6} f(n)$. Both overheads in (5) are at most $c_0 f(n)$ for a fixed $c_0$. Choose the induction threshold above 16, where the recurrence holds, and large enough that both recursive indices are at least 2 and strictly less than $n$. Choose $C \ge 8c_0$, increased to cover the finitely many values below the threshold. Strong induction then gives $T(n) \le Cf(n)$: the two bounds are $(7C/8 + c_0)f(n)$ and $(5C/6 + c_0)f(n)$, both at most $Cf(n)$. Finally take $n = \lceil \log a \rceil$. For $a \ge 4$, $n \le 2 \log a$ and $\log n \ge \log \log a \ge 1$, so $G(a, b) \le T(n) \le 2C \log a / \log \log a$. □

### 4.1 The leading coefficient

The same construction allows a more precise bound without changing either arithmetic lemma.

**Corollary 4.** *For every $\varepsilon > 0$, there is an $a_0$ such that for every $a \ge a_0$ and $0 \le b \le a$,*

$$G(a, b) \le (6 + \varepsilon) \frac{\log a}{\log \log a}.$$

*Equivalently, the uniform upper bound has coefficient $6 + o(1)$.*

*Proof.* Fix $0 < c < 1/2$ and $0 < \beta < 1/2$, and put $\alpha = 1 - \beta/2$. For large $n$ put

$$p = \lfloor c \log n \rfloor, \qquad k = 2^p, \qquad h = k^2 + 1, \qquad J = \lceil n/p \rceil.$$

Use $N^\alpha$ as the early threshold and $N^\beta$ as the target modulus, where $N = 2^n$. In the late case $r_h > N^\alpha$, so $2a/r_h < 2N^{1-\alpha} = 2N^{\beta/2} < N^\beta$ for sufficiently large $n$, and the prefix lemma applies to every modulus above $N^\beta$. Since $k^2 < h$, the prefix occupies at least $k+1$ residue classes, so each contraction reduces the modulus by a factor greater than $k$. Moreover $k^J \ge N$, so $J$ contractions suffice. Writing $f(n) = n/\log n$, we have

$$h = O(n^{2c}) = o(f(n)), \qquad 3J = \left( \frac{3}{c} + o(1) \right) f(n).$$

The two branches therefore give

$$T(n) \le \max \left\{ T(\lceil \alpha n \rceil) + o(f(n)), \quad 2T(\lceil \beta n \rceil) + \left( \frac{3}{c} + o(1) \right) f(n) \right\}.$$

Theorem 1 implies that $\Lambda = \limsup_{n \to \infty} T(n)/f(n)$ is finite and nonnegative. For each fixed $q > 0$, the ratio $f(\lceil qn \rceil)/f(n)$ tends to $q$. Taking upper limits in the recurrence yields

$$\Lambda \le \max\{\alpha\Lambda,\ 2\beta\Lambda + 3/c\}.$$

Since $\alpha < 1$ and $\Lambda \ge 0$, this implies $(1 - 2\beta)\Lambda \le 3/c$. First let $\beta \downarrow 0$ and then $c \uparrow 1/2$ to obtain $\Lambda \le 6$. Finally, $n = \lceil \log a \rceil$ satisfies $f(n) \sim \log a / \log \log a$, proving the assertion. All upper limits concern the worst-case function $T$, so the resulting threshold is independent of $b$. □

We do not claim that 6 is optimal.

### 4.2 Closure depth

Van den Dries and Moschovakis [10] measure the parallel depth of the gcd. Put $S_0 = \{0, 1, a, b\}$, let $S_{k+1}$ adjoin to $S_k$ all values $x+y$, $x-y$, $\mathrm{iq}(x,y)$ (integer quotient) and $\mathrm{rem}(x,y)$ with $x, y \in S_k$, and let $g(a,b)$ be the least $k$ with $\gcd(a,b) \in S_k$ [10, p. 2:6]. They ask whether there are positive integers $\alpha_n \to \infty$ and $\beta_n$ and a $C > 0$ with $g(\alpha_n, \beta_n) > C \log \alpha_n$ for all $n$ [10, Open Problem, p. 2:11].

**Corollary 5.** *For integers $\alpha \geq 4$ and $\beta \geq 1$,*

$$g(\alpha, \beta) \leq 1 + G(\alpha, \mathrm{rem}(\beta, \alpha)) = O\left(\frac{\log \alpha}{\log \log \alpha}\right).$$

*Hence no such sequences exist.*

*Proof.* Put $r = \mathrm{rem}(\beta, \alpha)$, so $r \in S_1$, $0 \leq r < \alpha$ and $\gcd(\alpha, r) = \gcd(\alpha, \beta)$. In a derivation from $\alpha, r$, the value of the $k$th instruction lies in $S_{k+1}$, by induction on $k$: its operands are $\alpha$, $r$ or earlier values, all in $S_k$. The source's remainder agrees with ours at positive divisors. Theorem 1 bounds $G(\alpha, r)$. □

## 5 Certificates and uniform processes

**Structures and certificates.** Write $\mathrm{tt}, \mathrm{ff}$ for the truth values and $\mathrm{eq}_c$ for the test $x = c$. We use the book's partial remainder operation, defined only when its second argument is nonzero, in the structures

$$\mathcal{N}_0 = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0), \qquad \mathcal{N}_\varepsilon = (\mathbb{N}, \mathrm{rem}, \mathrm{eq}_0, \mathrm{eq}_1).$$

The latter is the book's Euclidean structure $\mathbf{N}_\varepsilon$ [6, (1C-10), p. 24]; our $\mathcal{N}_0$ should not be confused with the book's $\mathbf{N}_0$, which has no remainder operation [6, p. 171]. Coprimeness is the relation $x \perp\!\!\!\perp y \iff x, y \geq 1 \wedge \gcd(x, y) = 1$ [6, §1A].

A derivation records finitely many remainder equations. These form a partial structure generated by the inputs. A *certificate* for $f$ at $\vec{x}$ in a structure $\mathcal{A}$ is such a finite generated partial substructure, possibly with test facts, on which every homomorphism into $\mathcal{A}$ respects $f$ at $\vec{x}$. In other words, any assignment of values preserving the recorded operations and tests must preserve the answer. The *intrinsic remainder-call complexity* $\mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, f, \vec{x})$ is the least number of remainder facts in a certificate [6, §§1D, 4C–4E].

The Lean development represents a certificate by its facts. The following lemma justifies this, and supplies the derivations used below.

**Lemma 6** (Fact representation)**.** *Let $F$ be a finite set of true remainder and test facts in which every number is generated from $a, b$ by the remainder facts of $F$. Then $F$ is the diagram of a finite partial substructure generated by $a, b$ (of $\mathcal{N}_\varepsilon$, or of $\mathcal{N}_0$ if $F$ has no $\mathrm{eq}_1$ facts), and every such substructure arises in this way. A map $\pi : \mathbb{N} \to \mathbb{N}$ preserves the facts of $F$ exactly when its restriction to the universe is a homomorphism, and every homomorphism extends to such a map. Every element of the universe is available after a derivation from $a, b$ with at most as many instructions as $F$ has remainder facts.*

*Proof.* Take the universe $V$ to consist of $a, b$ and the numbers occurring in $F$, and interpret each primitive only at the recorded arguments, with the recorded value. Truth of the facts makes these partial operations and tests restrictions of the primitives of $\mathbb{N}$, and the hypothesis says that $V$ is generated by $a, b$. Conversely, the diagram of a finite generated partial substructure is such a set

$F$. A homomorphism need only respect the recorded operations and tests on $V$, and values outside $V$ are irrelevant; this gives the second assertion. For the last, start from $a, b$ and repeatedly apply a remainder fact of $F$ whose operands are available and whose value is new. While some element of $V$ is unavailable such a fact exists: take the first unavailable value in a generation of that element. Each fact is applied at most once. □

**Proposition 7.** *For coprime $a, b \geq 1$, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) = D(a, b)$. For $(a, b) \neq (0, 0)$,*

$$G(a, b) \leq \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0, \gcd, a, b) \leq G(a, b) + 2,$$

*and the same bounds hold for $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd, a, b)$.*

*Proof.* A derivation of 1, together with $\mathrm{eq}_1(1) = \mathrm{tt}$, $\mathrm{eq}_0(a) = \mathrm{ff}$ and $\mathrm{eq}_0(b) = \mathrm{ff}$, certifies $\perp\!\!\!\perp$: under any homomorphism the input images are positive, and each of their common divisors divides every transported generated value, hence divides the image 1. Conversely, if a certificate has no positive $\mathrm{eq}_1$ fact, then $x \mapsto 2x$ preserves its remainder facts, zero tests and negative one-tests, but not coprimeness. So every certificate contains 1, and Lemma 6 gives a derivation of 1 with at most as many instructions as the certificate has remainder facts. This proves the first assertion.

For gcd as a function, the certificate must reach its output $g$, giving a lower bound of $G(a, b)$. For an upper bound, append

$$\mathrm{rem}(a, g) = z, \qquad \mathrm{rem}(b, g) = z, \qquad \mathrm{eq}_0(z) = \mathrm{tt},$$

where the generated $z$ is zero. Every homomorphism $\pi$ has $\pi(z) = 0$ and $\pi(g) > 0$ (by definedness of either appended remainder). Thus $\pi(g)$ divides both input images. Conversely every common divisor of those images divides $\pi(g)$ by induction over the derivation. This proves the bounds for $\mathcal{N}_0$. The lower-bound argument applies to every gcd certificate, $\mathrm{eq}_1$ facts included, and allowing $\mathrm{eq}_1$ facts can only lower the count, so the upper bound transfers to $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd, a, b)$. □

*Remark* 8. At $(0, 0)$ the output is an input, and the substructure with universe $\{0\}$ and no facts is a certificate. For gcd restricted to positive inputs, appending $\mathrm{eq}_0(a) = \mathrm{ff}$ and $\mathrm{eq}_0(b) = \mathrm{ff}$ keeps input images in the domain, at no remainder cost. For a positive noncoprime input, adding $\mathrm{eq}_1(g) = \mathrm{ff}$ to the gcd certificate certifies noncoprimeness, so the sublogarithmic remainder-call bound holds for both truth values of coprimeness as well as for the gcd. The overhead is a fixed additive constant and does not change the leading coefficient, so Corollary 4 holds for all three complexities of Corollary 10 below on positive inputs.

**A single uniform process.** Proposition 7, Remark 8, and Theorem 1 bound each intrinsic complexity by $O(\log a / \log\log a)$, uniformly for $a \geq b \geq 1$. The book's Theorem 4F.1 states that, whenever every input has a certificate, one uniform process attains the intrinsic complexity at every input [6, pp. 149–150]. Hence:

**Corollary 9.** *For each of gcd and coprimeness, there is a single uniform process of $\mathcal{N}_\varepsilon$ computing it on $\mathbb{N}^2$ with $O(\log a / \log\log a)$ remainder calls, uniformly for $a \geq b \geq 1$. The gcd process can also be taken over $\mathcal{N}_0$.*

*Proof.* The certificate bounds give the cost assertion by Theorem 4F.1. Every input has a certificate: symmetry covers reversed inputs, and a zero-test fact suffices when either input is zero. Thus the same process is defined on all of $\mathbb{N}^2$. □

The corollary rules out the possibility raised after Corollary 6C.7 of the book that the Main Conjecture holds for all uniform processes of $\mathcal{N}_\varepsilon$ [6, p. 188].

**The conjectures refuted.** Moschovakis's 2012 notes state:

> **Conjecture 3.4** [5, p. 19]. There is a rational number $r > 0$ such that for infinitely many $a \geq b \geq 1$, $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) \geq r \log a$.

The book formulates one precise version of the Main Conjecture [6, p. 151]: there are a set $\mathcal{D} \subseteq \mathbb{N}^2$ and a $K > 0$ such that

$$(x, y) \in \mathcal{D} \implies \mathrm{calls}_{\mathrm{rem}}(\mathrm{Eucl}, x, y) \leq L(x, y) \leq K \, \mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, x, y), \tag{4F-5}$$

where $L(x, y) = 2 \log \max(x, y)$ for $(x, y) \neq (0, 0)$, $L(0, 0) = 0$, and $\mathrm{calls}_{\mathrm{rem}}(\mathrm{Eucl}, x, y)$ counts the remainder calls of the Euclidean algorithm Eucl (the book writes $\varepsilon$) on $(x, y)$. The book states (4F-5) for the Euclidean algorithm in $\mathbf{N}_\varepsilon$, so its $\mathcal{A}$ is $\mathcal{N}_\varepsilon$; we treat both $\mathcal{A} = \mathcal{N}_\varepsilon$ and $\mathcal{A} = \mathcal{N}_0$. The book adds that $\mathcal{D}$ then satisfies its conditions (4F-3) and (4F-4); the latter makes the intrinsic complexity unbounded on $\mathcal{D}$, so $\mathcal{D}$ is infinite (for finite $\mathcal{D}$ the statement is trivial). We refute the weaker statement obtained by dropping the first inequality and requiring only that $\mathcal{D}$ be infinite.

**Corollary 10.** *The intrinsic complexities*

$$\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b), \qquad \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_0, \gcd, a, b), \qquad \mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \gcd, a, b)$$

*are $O(\log a / \log \log a)$, uniformly for $a \geq b \geq 1$. Consequently Conjecture 3.4 of [5] is false, and so is the version (4F-5) of the Main Conjecture with $\mathcal{A} = \mathcal{N}_0$ or $\mathcal{A} = \mathcal{N}_\varepsilon$: for every $r > 0$ only finitely many pairs $a \geq b \geq 1$ satisfy $\mathrm{calls}_{\mathrm{rem}}(\mathcal{N}_\varepsilon, \perp\!\!\!\perp, a, b) \geq r \log a$, and for every $K > 0$ only finitely many pairs $(x, y) \in \mathbb{N}^2$ satisfy $L(x, y) \leq K \, \mathrm{calls}_{\mathrm{rem}}(\mathcal{A}, \gcd, x, y)$.*

The corollary follows from Theorem 1 and Proposition 7; for noncoprime pairs, the bound on the coprimeness complexity comes from Remark 8. For (4F-5), pairs with $x < y$ or a zero entry are handled as in the proof of Corollary 9; the latter need no remainder facts.

**The limit of the homomorphism method.** The Homomorphism Test proves lower bounds on intrinsic complexity by showing that every generated diagram below the proposed cost admits a homomorphism that fails to preserve the answer [6, Lemma 4E.3]. Such bounds apply to every uniform process computing the function, hence to every recursive program [6, Proposition 4E.1, Corollary 4E.2]. By Corollary 10, no lower bound on the intrinsic complexities listed there can grow faster than $\log a / \log \log a$, so a logarithmic lower bound for algorithms must use restrictions beyond uniformity, such as finite program syntax or determinism.

## 6 An almost-all lower bound

The difficult pairs of [6, Theorem 6C.5] give $\Omega(\log \log a)$ on a sparse set. An elementary Diophantine argument gives the same order for almost all coprime pairs. Write each generated value as $ua + wb$. On the $B$-regular pairs defined below, small coefficients cannot give a small positive value, and each remainder call at most squares the coefficient bound, up to a factor $2B$. Reaching 1 therefore takes about $\log \log a$ calls, except on a sparse set of pairs.

**Proposition 11.** *For integers $d \geq 1$ and $N \geq 3$, at most $40(2 + \ln N) N^{2 - 2^{-d}}$ coprime pairs $2 \leq b < a \leq N$ satisfy $D(a, b) \leq d$. Hence, for each fixed $\varepsilon > 0$, all but $o(N^2)$ of these pairs satisfy $D(a, b) > (1 - \varepsilon) \log \log N$.*

*Proof.* For $B \geq 1$ call $(a, b)$ *$B$-regular* if $|qb - pa| \geq a/(Bq)$ whenever $q \in \mathbb{N}$, $q \geq 1$, $p \in \mathbb{Z}$ and $qb \neq pa$. We first bound the cost on regular pairs, then count the irregular ones.

Every generated value has an integer representation $v = ua + wb$. On a regular pair, $v > 0$ implies $v \geq a/(B(|u| + |w|))$: for $w \neq 0$ use regularity with $q = |w|$, and for $w = 0$ note $v \geq a$. Let $H_t$ bound the coefficient norm $|u| + |w|$ in one such representation of each value available after $t$ instructions, starting with $H_0 = 1$. In an instruction $r = \mathrm{rem}(x, y)$, write $x = sy + r$. Since generated values are at most $a$, the quotient satisfies $s \leq a/y \leq BH_t$. Subtracting $s$ times the representation of $y$ from that of $x$ gives

$$H_{t+1} \leq H_t + BH_t^2 \leq 2BH_t^2, \qquad H_t \leq (2B)^{2^t - 1}.$$

Producing 1 within $d$ instructions therefore forces

$$1 \geq \frac{a}{BH_d}, \qquad a \leq \frac{(2B)^{2^d}}{2}.$$

To count irregular pairs below $N$, choose a witness $0 < |qb - pa| < a/(Bq)$. Dividing $p, q$ by their gcd preserves this inequality, so assume $\gcd(p, q) = 1$. The nonzero integer $r = qb - pa$ then gives $q < a/B \leq N/B$ and $0 \leq p \leq q$. For each $q$, there are at most $2q$ choices of $p$ and $2N/(Bq)$ choices of the signed integer $r$. Given $p, q, r$, the equation $pa \equiv -r \pmod{q}$ determines one residue class for $a$, leaving at most $N/q + 1$ choices; $b$ is then determined. With $Q = \lfloor N/B \rfloor$, the number of irregular pairs is at most

$$\begin{aligned}\sum_{q=1}^{Q} (2q) \frac{2N}{Bq} \left( \frac{N}{q} + 1 \right) &= \frac{4N^2}{B} \sum_{q=1}^{Q} \frac{1}{q} + \frac{4NQ}{B} \\ &\leq \frac{4N^2}{B} (2 + \ln N),\end{aligned}$$

using $Q \leq N$, $B \geq 1$, and the harmonic-sum bound; an empty sum is zero.

Now split into dyadic ranges $X/2 < a \leq X$, and take $B = X^{2^{-d}}/2$. If $B \geq 1$, a regular pair with $D \leq d$ would have $a \leq X/2$, so all such pairs in the range are irregular. Their number is at most $8(2 + \ln X) X^{2 - 2^{-d}}$. When $B < 1$, the same bound follows from the trivial bound $X^2$, since $X^{2^{-d}} < 2$. Put $s = 2 - 2^{-d} \geq 3/2$. Summing over the nonempty ranges with $X = N/2^j \geq 3$, and bounding their logarithmic factors by $2 + \ln N$, gives

$$8(2 + \ln N) N^s \sum_{j \geq 0} 2^{-js} = \frac{8}{1 - 2^{-s}} (2 + \ln N) N^s \leq 40(2 + \ln N) N^s.$$

For the second assertion, take $d = \lfloor \log \ln N - 2 \log \ln \ln N \rfloor$ for sufficiently large $N$. Then $N^{-2^{-d}} \leq \exp(-(\ln \ln N)^2)$, so the exceptional set has size $o(N^2)$, while $d = (1 - o(1)) \log \log N$, so eventually $d \geq (1 - \varepsilon) \log \log N$. □

A positive proportion (asymptotically $6/\pi^2$) of these pairs are coprime, so the exceptional set also has density zero among coprime pairs. The saving $2^{-d}$ in the exponent of $N^{2 - 2^{-d}}$ shrinks as $d$ grows, so this argument does not reach the upper-bound scale $\log N / \log \log N$.

## 7 Fixed programs with arbitrarily small coefficients

For a nondeterministic program $P$, let $c_P(a, b)$ be the minimum number of remainder calls among its convergent computations. We require at least one such computation on each input and require every convergent computation to return the gcd. These are the conventions of [6, (2E-2), (2E-5)].

A *register program* keeps its values in a fixed number of registers and executes assignments $x_l := E$, branches on a test $E$, and a halt returning $E$, where each $E$ is a term in the registers; the nondeterministic version allows several commands at a line [6, §§2E–2F]. It is an iterator, so it uses no recursion stack, and in pointed structures register programs compute exactly the tail-recursive functions [6, Proposition 2F.1].

**Theorem 12.** *For each integer $p \geq 1$, there is a finite nondeterministic register program $P_p$ over $\mathcal{N}_0$ with $4^p + 7$ registers that computes the gcd and satisfies*

$$\limsup_{a\to\infty} \max_{1\leq b\leq a} \frac{c_{P_p}(a,b)}{\log a} \leq \frac{3}{p}.$$

*Consequently, for every $\varepsilon > 0$ there are a fixed register program $P$ and a constant $B \geq 0$ such that*

$$\forall a \geq b \geq 1, \qquad c_P(a,b) \leq \varepsilon \log a + B.$$

The program depends on $\varepsilon$ but not on the inputs. The Main Conjecture lets its coefficient $r$ depend on the program, so the family is consistent with it; no single sublogarithmic program is obtained.

*Proof.* Put $k = 2^p$, $h = k^2 + 1$ and $W = h + 6$. If $b = 0$, return $a$. Otherwise initialise the registers $x_0, \ldots, x_{W-1}$ at $(a, b, a, b, a, \ldots, a)$ and enter the main line. There the program may perform any assignment $x_l := \operatorname{rem}(x_i, x_j)$ with $i, j, l \geq 2$, or pass to verification with a candidate $x_i$, $i \geq 2$. Verification halts with $x_i$ if $x_i \neq 0$, $\operatorname{rem}(x_0, x_i) = 0$ and $\operatorname{rem}(x_1, x_i) = 0$, and loops otherwise. The inputs $x_0, x_1$ are never overwritten. Every register value is divisible by $\gcd(a, b)$, so a candidate that passes verification is the gcd, and Euclid's algorithm gives a convergent computation. The computations below move values between registers for readability; copying is never needed (Appendix A).

Multiplying every value by $g = \gcd(a, b)$ turns a computation on $(a/g, b/g)$ into one on $(a, b)$ with the same remainder count, so it suffices to treat coprime pairs. Let $S(n)$ be the largest, over coprime $1 \leq b \leq a \leq 2^n$, of the least number of remainder calls that, starting from any state with $a, b$ in two working registers, place 1 in a working register. Fix $0 < \beta < 1/2$ and put $\alpha = 1 - \beta/2$ and $N = 2^n$.

Compute the Euclidean remainders $r_0 = a, r_1 = b, r_2, \ldots$ in working registers, stopping at $r_h$ or at the first $r_j \leq N^\alpha$. If 1 has been reached, no further generation is needed. In the remaining early case, one further remainder gives the positive coprime pair $r_j, r_{j+1}$, bounded by $N^\alpha$, at total cost at most $h$; continue with that pair. Otherwise $r_h > N^\alpha$. As in the proof of Corollary 4, now with $p$ fixed, the prefix lemma applies to every modulus above $N^\beta$ for large $n$, the prefix occupies at least $k + 1$ residue classes, and at most $\lceil n/p \rceil$ contractions, using a modulus register and two scratch registers, produce a positive $m \leq N^\beta$. Then put $u = a \bmod m$ and $v = b \bmod m$, and run Euclid's algorithm in place on $m, u$ and then on the result $d$ and $v$. Each run takes at most $2 \log m + 3$ calls, and the result is $\gcd(d, v) = \gcd(m, a, b) = 1$. Thus, eventually in $n$,

$$S(n) \leq \max\left\{ S(\lceil \alpha n \rceil) + h, \quad h + 3\lceil n/p \rceil + 4\beta n + 7 \right\}.$$

Euclid gives $S(n) = O(n)$. Passing to upper limits, $\Lambda = \limsup_{n\to\infty} S(n)/n$ satisfies $\Lambda \leq \max\{\alpha\Lambda,\ 3/p + 4\beta\}$, and since $\alpha < 1$, $\Lambda \leq 3/p + 4\beta$. The program does not depend on $\beta$, so $\Lambda \leq 3/p$. Take $n = \lceil \log a \rceil$ and add the two verification calls. Finally, choose $p > 3/\varepsilon$ and absorb the finitely many smaller inputs in $B$. □

**Recursive programs.** With the prefix length fixed, Euclid's algorithm completes the computation at cost $O(\beta n)$, negligible as $\beta \to 0$; the recursive completion of Section 4 is needed only when the prefix length grows. Writing one function per line turns the register program into a finite nondeterministic tail-recursive program in the sense of [6, §§2A, 2E], as in the book's reduction of iteration to tail recursion [6, §2C]: a function run of all the registers, with one body for each assignment, calling run with $\mathrm{rem}(x_i, x_j)$ in place of $x_l$, and one for each candidate, verifying $x_i$. Lean checks that its evaluations, and its computations on a specialisation of the book's recursive machine, correspond exactly to the computations of the register program, with the same remainder cost (Appendix A), and that the machine's stacks stay bounded: at most $\max(4^p + 11, 17)$ entries in all, on every input and however long the computation runs. So Theorem 12 holds for the book's recursive programs as well, with no growing recursion stack.

## 8 Discussion and open problems

**The gap.** The intrinsic complexity of coprimeness in $\mathcal{N}_\varepsilon$ is at least $\Omega(\log \log a)$ on almost all coprime pairs and at most $O(\log a / \log \log a)$ on all positive pairs. Closing this gap is open; unlike Bshouty's bound in his model, our upper bound is not known to be optimal.

**One program.** The remaining obstacle to a single sublogarithmic program is reusable access to a prefix whose length grows with the input. Selecting a close pair of residues costs nothing in a derivation, whereas computing all prefix residues would take about $h$ calls per contraction. By Theorem 12, $W = 4^p + 7$ registers give coefficient $3/p$, roughly $6/\log W$, so no bound $c \log a$ with $c$ independent of the number of registers holds for nondeterministic register programs. A register program has only a fixed number of slots for prefix values. Recursion permits unbounded saved state, but does not directly provide reusable access to an arbitrarily long prefix. Does every nondeterministic register program with $W$ registers make at least $c \log a / \log W$ remainder calls on infinitely many inputs, for an absolute $c > 0$? Can one fixed recursive program retain and repeatedly access a growing prefix at the same remainder cost? A lower bound cannot rest on how quickly values shrink, since $\mathrm{rem}(m + 1, m) = 1$, and Proposition 2E.1 of the book [6, §2E] converts each certificate into a computation of some program, not of one program for all certificates. Pratt's algorithm is itself a nondeterministic register program [6, Theorem 2E.2]; compare also the book's question whether recursion reduces to tail recursion in the pointed Euclidean structure [6, Problem x2G.6].

**Stronger lower-bound methods.** The limit established in Section 5 leaves room for arguments combining homomorphisms with additional restrictions on computations. In particular, the conjecture of a logarithmic lower bound for deterministic uniform processes remains open [5, Conjecture 3.9, p. 23], [6, §4H and p. 188].

## 9 Computational checks

The Python planner `remainder_only_certificate.py`, in the repository https://github.com/jasondavies/euclid-uniform-processes, chooses instructions offline with unrestricted arithmetic, using the parameters of Section 4; below a base size (256 bits by default) it uses Euclid's algorithm. An independent replay then checks operand availability, nonzero divisors, every remainder equation and the final gcd.

The script `scripts/benchmark_certificates.py` uses seed 20261016 to sample 25 coprime pairs at each size: $a$ is uniform among integers of that bit length, and $b$ is drawn uniformly from $\{2, \ldots, a-1\}$ until $\gcd(a, b) = 1$. It also runs the successive Fibonacci pair $(F_k, F_{k-1})$ with $F_k$ the first Fibonacci number of that bit length ($k = 738$, 2951, 11801). The table gives remainder calls; the Euclidean count stops when 1 first appears.

| | Random pairs: min / median / max | | Fibonacci pair | |
|---|---|---|---|---|
| Input bits | Construction | Euclid | Construction | Euclid |
| 512 | 189 / 214 / 226 | 266 / 295 / 337 | 38 | 735 |
| 2048 | 617 / 647 / 669 | 1136 / 1192 / 1260 | 60 | 2948 |
| 8192 | 2007 / 2037 / 2059 | 4664 / 4774 / 4890 | 110 | 11798 |

All 78 derivations and the example of Section 3 pass replay, and `validation/benchmark_certificates.json` records the individual counts. On the Fibonacci pairs, Euclid's worst case, the bits removed by successive contractions roughly double (16, 31, 64, 125 and 252 at 512 bits), so few contractions are needed. These samples illustrate the construction; the asymptotic claim rests on the proof. The accompanying repository also checks scaled and noncoprime inputs, rejects invalid traces, and exhaustively tests the prefix lemma over a finite range. The `README.md` records commands, test ranges, seeds, and results. The Python checks are not formally verified; the unrestricted arithmetic used to plan instructions is outside the derivations.

# A Formal verification

The development uses Lean 4.35.0 [2] with mathlib [4] at its v4.35.0 tag; the pinned mathlib commit is `56a005855ea17e0b0958f73ec801ae140b0a1679`. The proved library contains no proof placeholders or added axioms; the main-result audit reports only `propext`, `Classical.choice`, and `Quot.sound`. Table 1 gives the main declarations; the repository README gives the complete map and reproduction commands. The general theory of uniform processes, including Theorem 4F.1 used in Corollary 9, and the historical lower bounds are cited, not formalised. The certificate bounds and conjecture refutations are formalised independently of that theory, with certificates represented by their facts as in Lemma 6; that lemma and Corollary 5 are proved here, not formalised.

The development is also packaged for the Palomar registry: `Challenge.lean` imports only mathlib and states the main theorems with every definition visible and deliberate `sorry` placeholders. `Solution.lean` imports the proved library, independently of these placeholders. Palomar's comparator checks that the proved statements are exactly those of `Challenge.lean` and replays the proofs through Lean's kernel and independent kernels. The thirteen compared theorems are marked † in Table 1. The registry entry [7] records this verification for commit `40279dd`.

**Arithmetic bounds.** Costs are lengths of explicit remainder derivations. The finite theorem `sizeCost_step` proves the recurrence from natural-number size parameters; it does not assume a recurrence for the cost function. For the initial Big-O proof, Lean uses $j = \lfloor \log n \rfloor$, $p = \lfloor j/4 \rfloor$ and $h = 2^{2p} + 1$ in place of $h = \lceil \sqrt{n} \rceil$, and checks the integer rounding estimates before passing to real logarithms. The refinement uses the tunable parameters of Corollary 4 and checks the rounding limits preserving coefficient 6. For Proposition 11, Lean takes $B = a^{2^{-d}}/2$ separately for each pair instead of splitting into dyadic ranges. It proves the stated constant 40 and the exceptional-set limit for every fixed $\varepsilon > 0$, with threshold $(1 - \varepsilon) \log \log N$.

Table 1: Main results and Lean declarations, all in the namespace `ShortGCD`. Names marked † are compared against `Challenge.lean`.

| Paper | Lean declaration(s) |
|---|---|
| Theorem 1 | `uniform_real_bound`, `sizeCost_isBigO` |
| Normalisation (1) | `G_normalize` |
| Lemma 2 | `small_remainder` |
| Lemma 3 | `prefix_pairs_distinct`, `residue_card_sq` |
| Corollary 4 | `uniform_real_bound_six`†, `uniform_intrinsic_bound_six`† |
| Proposition 7 | `callsRem_eq_D`, `G_eq_D`, `callsCoprimeness_eq_callsRem`, `gcd_certificate_lower`, `callsGcd_le`, `callsRemGcd_le_callsGcd`, `exists_certifiesGcd`†, `G_le_callsRemGcd`†, `G_le_callsGcd`†, `callsCoprimeness_eq_G`† |
| Remark 8 | `callsCoprimeness_le`, `exists_certifiesCoprimeness`† |
| Corollary 9 | Cited Theorem 4F.1; not formalised here |
| Corollary 10 | `not_conjecture_3_4`†, `not_main_conjecture_4F5`†, `not_main_conjecture_4F5_Neps`†; literal versions `not_conjecture_3_4_literal`†, `not_main_conjecture_4F5_literal`† |
| Proposition 11 | `AlmostAll.card_D_le`, `AlmostAll.tendsto_card_within_loglog`, `tendsto_card_callsCoprimeness_loglog`† |
| Theorem 12 | `TailProgram.maxCost_step`, `TailProgram.Fixed.limsup_profile_le`, `TailSource.run_trace_relabel`, `TailSource.executes_correct`, `TailSource.arbitrarily_small_coefficient` |
| Recursive programs (Section 7) | `TailRecursive.program_tailRecursive`, `TailRecursive.stacks_bounded`, `TailRecursive.executes_iff_eval`, `Recursive.Machine.eval_iff_steps`, `TailRecursive.arbitrarily_small_coefficient` |

**Programs and machine computations.** For Theorem 12, Lean defines the register program itself (`TailSource.Step`): finitely many lines, work registers, two input registers that no command writes, and no stack. Each command is an assignment $x_l := \mathrm{rem}(x_i, x_j)$ or a branch or halt on a single register, and each remainder evaluation is charged once, including the two verification calls. Every convergent execution returns the gcd, and one exists on every input. The cost proofs use a generator that may also relabel all registers at once. `TailSource.run_trace_relabel` replays such runs by single-register commands at the same cost and with no extra registers: it tracks the relabelling in the proof, and writes each new remainder to a register no other value occupies, which exists by pigeonhole whenever two names share a register.

The tail-recursive form (`TailRecursive`) is written in a language of finite nondeterministic first-order programs (`Recursive`): terms with variables, remainder, a zero-test conditional and calls of internal symbols, each symbol with finitely many bodies, evaluated call by value. `TailRecursive.program_tailRecursive` certifies that the head and every body have only tail-position internal calls: arguments, remainder operands and conditional guards contain no internal calls. Consequently (`TailRecursive.stacks_bounded`) the machine's control and value stacks never hold more than $\max(w+6, 17)$ entries in all, for $w$ work registers, on any input and however long the computation runs; each closure environment has at most $w+2$ entries. `TailRecursive.executes_iff_eval` matches its evaluations exactly, in value and remainder cost, with executions of the register program. For every program in the language, `Recursive.Machine.eval_iff_steps` proves exact-cost equivalence with a specialisation of the book's two-stack machine: arguments are evaluated right to left, zero tests and branches are separate transitions, and only remainder transitions are charged [6, Table 1, pp. 75–76]. Closures represent substitution in machine states, not extra program data.

## Declaration of AI-generated research and authorship

This manuscript is the product of an AI-driven research process. During a broader investigation initiated by the human author, an AI system, GPT-6-Astra, identified the principal result. The author then directed AI systems to formalise the results in Lean and to prepare this manuscript. The AI systems developed the results and their presentation, including the mathematical exposition, the formalisation, and the organisation and writing of the text; the human author wrote no part of the manuscript. The systems were GPT-6-Astra (OpenAI), used through Codex, and Claude Opus 5.5 (Anthropic), used through Claude Code. Both contributed to the Lean formalisation and to the review and revision of the manuscript.

The Lean development gives a machine-checkable version of the main claims. Lean's kernel checks the proofs, including the mathlib results they use, relative to the axioms listed in Appendix A. The correspondence between the formal definitions and the book's notions is a separate mathematical obligation (Lemma 6).

This declaration distinguishes the provenance of the work from conventional AI-assisted writing. The human author accepts responsibility for the submission and for the accuracy of this disclosure.

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